

Homework 1

ASEN 5090 Introduction to GNSS

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Problem 1

(a)

$$\text{PL1} = 550 \text{ [m]}$$

$$\text{PL2} = 500 \text{ [m]}$$

Figure 1: Insert figure caption here.

Let

$$\rho^k = \sqrt{(x^k - x)^2 + (y^k - y)^2 + (z^k - z)^2} + b.$$

Given the setup, this can be constrained to a 1D problem on the x axis,

$$\rho^k = \sqrt{(x^k - x)^2} + b.$$

Which is equivalent to,

$$\rho^k = |(x^k - x)| + b.$$

Given the setup on a line, we have the ρ^1 , the observed position be $550m$, and x^1 position on the line being 0 . Likewise $\rho^2 = 500m$ and $x^2 = 1000m$. This gives a set of equations

PL1:

$$550m = |0 - x| + b$$

$$550m = x + b$$

PL2:

$$500m = |1000m - x| + b$$

$$-500m = -x + b$$

This can be solved by elimination, eliminating the bias b giving,

$$1050m = 2x$$

$$\boxed{\therefore x = 525 \text{ [m]}}$$

Rewrite the PL1 equation to solve for b .

$$\boxed{\begin{aligned} b &= 550m - x \\ \therefore b &= 25 \text{ [m]} \end{aligned}}$$

htbp

Figure 2: Insert figure caption here.

b.

$$\text{PL1} = 400 [m]$$

$$\text{PL2} = 1400 [m]$$

The solution method remains the same, where we develop a system of equations from the observed ranging measurements and based on the known locations of the sat.

PL1:

$$400m = |x| + b$$

PL2:

$$1400m = |1000 - x| + b$$

And with eliminaton and eliminating x this time, we obtain,

$$800m = 2b$$

$$\therefore b = 400[m]$$

And solve for x with

$$x = b - 400$$

$$\therefore x = 0$$

Problem 2

(a)

From the diagram above, we are able to create solve for the following equations as functions of x_a, v_0, h_0 .

$$R(x_a, v_0, h_0)$$

$$\dot{R}(x_a, v_0, h_0)$$

$$z(x_a, v_0, h_0)$$

When solving for range, one can simply use the pythagorean theorem given h_0 is constant.

$$R = \sqrt{x_a^2 + h_0^2}$$

Solving for range rate, take the differential w.r.t time.

$$\begin{aligned}
 R &= \sqrt{x_a^2 + h_0^2} \\
 \dot{R} &= \frac{dR}{dt} \\
 &= \frac{1}{2} (x_a^2 + h_0^2)^{-1/2} (2x_a \dot{x}_a + 2h_0 \dot{h}_0) \\
 &= \frac{x_a \dot{x}_a + h_0 \dot{h}_0}{\sqrt{x_a^2 + h_0^2}}
 \end{aligned}$$

Since h_0 is constant, $\dot{h}_0 = 0$, and $\dot{x}_a = v_0$:

$$\dot{R} = \frac{x_a v_0}{\sqrt{x_a^2 + h_0^2}}$$

Solving for zenith, looking at the diagram, finding z as a function of x_a can simply be

$$z = \arctan\left(\frac{x_a}{h_0}\right)$$

(b)

See attached MATLAB script.

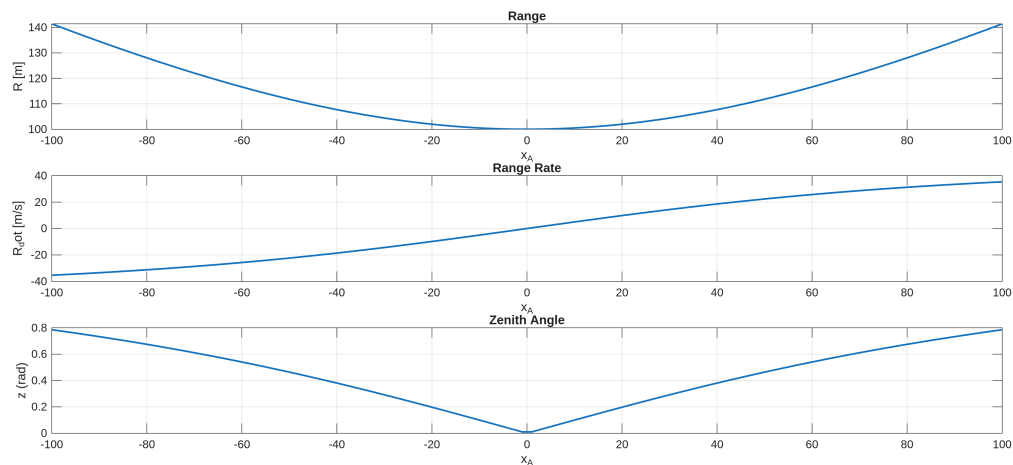


Figure 3: MATLAB Script Results

(c)

Write your solution here.

Problem 3

(a)

Write your solution here.

(b)

Write your solution here.

Problem 4

Write your solution here.